Chapter 12

Heterologous Expression of Candida Antifungal Target Genes in the Model Organism *Saccharomyces cerevisiae*

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Abstract

Understanding how a gene variant may influence antifungal resistance, or other phenotypic characteristics, is an important step in determining or dissecting resistance mechanisms. The influence of specific genes or gene alleles on a phenotype can initially be assessed within the model organism, *Saccharomyces cerevisiae*. *S. cerevisiae* exhibits efficient rates of homologous recombination making it amendable for heterologous expression and represents a susceptible organism that can be used to determine changes in antifungal susceptibilities. Many groups have developed different methodologies for the cloning, expression, and screening processes. In this chapter, we present straightforward methodology that utilizes gap-repair cloning to express a plasmid-borne copy of *Candida auris ERG11* within *S. cerevisiae*. Multiple alleles can be compared in order to determine how specific alterations influence triazole susceptibility. Primer design, gap-repair co-transformation, and colony PCR screening are detailed.

Key words *Candida auris*, *Saccharomyces cerevisiae*, Heterologous expression, Antifungals, Resistance, *ERG11*, Drug target cloning

1 Introduction

Discovery of cellular factors that may influence antifungal susceptibility or identification of new alterations within well-known target genes requires genetic manipulation and subsequent analysis to determine their role, or non-role, in antifungal resistance development. These studies inform the field as to how fungal cells respond to specific antifungals and which factors may warrant examination within clinical isolates.

While newer techniques of genetic manipulation (e.g., CRISPR) allow direct alteration of genes within a host organism, heterologous expression of gene(s) within *Saccharomyces cerevisiae* still stands as a straightforward and cost-effective method that can provide (1) initial insight into the significance of your gene or mutation of interest and (2) a workable system prior to the development or fine-tuning of more elaborate genetic manipulation

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techniques, particularly with respect to species that are less studied or difficult to genetically alter. Wild-type lab strains of *S. cerevisiae* are naturally susceptible to most antifungals and exhibit mechanisms of acquired resistance similar to multiple *Candida* species, making the organism a reliable model in the study of antifungal resistance. In addition, scientists can choose to work with either the diploid or haploid life form of *S. cerevisiae* depending on the *Candida* species that is being modeled.

Expression of *Candida* genes in *S. cerevisiae* has been used to determine the significance of many genes, including those that may influence drug susceptibility [1] or adherence and biofilm formation [2]. In addition, similar methodologies have been used to facilitate screening for inhibitors of fungal enzymes [3, 4]. Genes from medically relevant fungi other than *Candida* have also been expressed in *S. cerevisiae* [5].

Many *ERG11* mutations identified in triazole-resistant clinical isolates of *Candida albicans* were initially characterized through heterologous expression in *S. cerevisiae* [6]. *ERG11* encodes for lanosterol 14α-demethylase (Erg11p or CYP51), an enzyme within the ergosterol biosynthesis pathway and the triazole drug target. Follow-up studies that built several mutations directly into *C. albicans* confirmed the role of these mutations in triazole resistance [7]. These follow-up studies were directed by the initial findings within *S. cerevisiae*. More recently, we used this system to determine the significance of multiple *ERG11* mutations that were identified in clinical isolates of the emerging fungal pathogen, *Candida auris* [8, 9], which demonstrates exceedingly high levels of triazole resistance [10]. This was particularly useful at the start of our studies, as there were few genetic tools available for direct manipulation of *C. auris*. In a similar manner, we characterized *ERG11* mutations identified in susceptible dose-dependent and triazole-resistant clinical isolates of *Candida parapsilosis* [11].

There are multiple ways to express your gene of interest in *S. cerevisiae* including use of various cloning techniques and plasmids. Additionally, a recent study described the expression of *Candida* species *ERG11* in *S. cerevisiae* from a chromosome instead of from plasmid [4]. Specific variations within this methodology are based on the promoter type chosen (the gene can be under the control of a constitutive, inducible, or native promoter) and the methodology selected to determine proper gene expression. Regulation of gene and protein expression may differ in the model organism; therefore, one should consider which type of promoter might best suit the experimental goals. Expression can be confirmed through quantitative RT-PCR, immunofluorescence (if the genes are tagged) [2], western blotting, or phenotypically. In addition, the chosen plasmid must contain a selectable marker for the transformation process. Nutritional (e.g., *URA3*, *TRP1*, *HIS3*) and dominant drug resistance (e.g., *NAT*^{r}, Hyg^r) markers are commonly used. One more consideration is codon optimization. Specifically, most *Candida* species (*Candida* or CTG clade) translate the CUG codon as serine instead of leucine [12]. This is not true for non-CTG (*Saccharomyces* clade) fungi, including *S. cerevisiae*, *C. glabrata,* and *C. krusei* [13]. Therefore, if you plan to express a gene from a CTG clade fungus with either multiple CUG codons or CUG codons within sensitive (i.e., drug binding) parts of the produced protein, you must consider how insertion of leucine by *S. cerevisiae*, versus serine, may affect your results. If necessary, these codons can be altered.

In this protocol, we detail the cloning of *C. auris ERG11* and expression within *S. cerevisiae*. Our procedure utilizes plasmid pRS416 and one of the *S. cerevisiae* BY strains. However, this protocol can be adapted to express any fungal gene on your plasmid of interest. We describe the use of gap-repair methodology that allows direct transformation of *S. cerevisiae* and bypasses the ligation and transformation of *E. coli*. Since *S. cerevisiae* is proficient at homologous recombination, the transformation efficiency rates are manageable using gap repair (~80% of clones screened exhibit correct plasmid construct). If one wishes to perform site-directed mutagenesis for codon optimization or mutational analysis, plasmid DNA can be rescued from yeast cells and propagated in *E. coli* as previously described [14].

2 Materials

1. Purified genomic DNA from strain(s) of interest (here, *C. auris* isolates that contain either wild-type or specific Erg11 alterations) (*see* **Note 1**).
2. *S. cerevisiae* competent cells (ATCC 201388/BY4741: MATa his3Δ1 leu2Δ0 met15Δ0 ura3Δ0) (*see* **Note 2**).
3. Frozen EZ Yeast Transformation II kit (Zymo Research).
4. Plasmid pRS416 (*S. cerevisiae CEN/ARS*; *URA3*; AMPr) purified from *E. coli* ATCC 87521, if necessary (*see* **Note 3**).
5. SmaI restriction endonuclease and associated buffer (*see* **Note 4**).
6. Shrimp alkaline phosphatase (SAP).
7. Gap-repair primers to amplify gene of interest (*see* Methods and Fig. 1).
8. Additional PCR primers to screen transformants and sequence the gene insert (*see* Methods and Fig. 1).
9. PCR machine/thermocycler.
10. Other PCR materials: sterile water, dNTPs, high-fidelity polymerase like Phusion, enzyme buffer, PCR tubes.

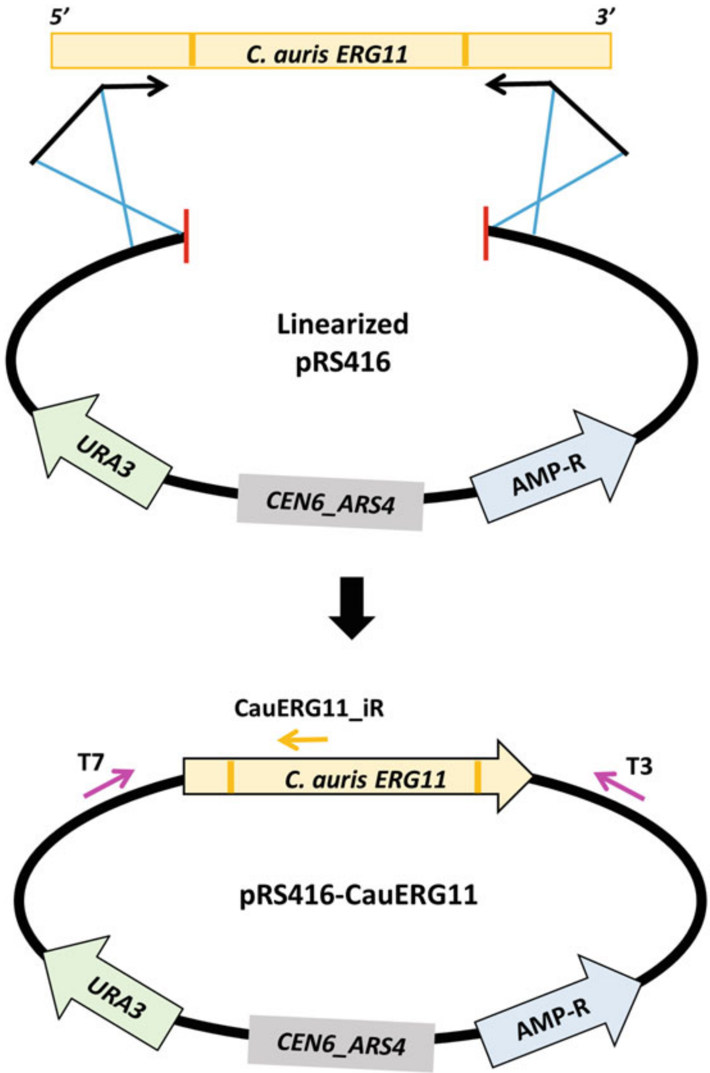

Fig. 1 Gap-repair primer design and cloning strategy. Primers can be designed to amplify the coding region alone, or coding region plus regulatory regions of your gene of interest. Here, *C. auris ERG11* coding and regulatory regions are amplified with *CauERG11*-specific primers that contain overhangs homologous to each side of the SmaI restriction site on pRS416. The 3′ ends of the primers (approximately 20 nucleotides) should adhere to your gene of interest, while the 5′ overhang regions (approximately 50 nucleotides) match to the plasmid. Competent *S. cerevisiae* cells are then co-transformed with the purified PCR product and alkaline phosphatase-treated and SmaI-linearized pRS416. Correct construct presence is confirmed through use of screening primers that adhere to the plasmid backbone (e.g., T7) and to the cloned gene (e.g., CauERG11_iR)

11. Stationary and shaking incubator (set at 30 °C).
12. PCR purification kit (similar to Qiagen QIAquick kit).
13. 1% w/v agarose gels and standard electrophoresis equipment.
14. DNA loading dye and DNA ladder.
15. Spectrophotometer (ideally microvolume, such as Nanodrop).
16. YPD broth: 10 g/L yeast extract, 20 g/L peptone, 20 g/L dextrose.
17. Selective growth medium, broth and agar plates. Here, SC-ura: 1.71 g/L Yeast Nitrogen Base (YNB), 5 g/L ammonium sulfate, 20 g/L glucose, CSM-ura (w/v on container), 18 g/L agar (for plates).
18. Benchtop centrifuge and microcentrifuge.
19. EmeraldAmp PCR mastermix (Takara Bio) (*see* **Note 5**).
20. Yeast DNA isolation kit (similar to Zymo Research Quick-DNA Fungal/Bacterial Miniprep Kit).

3 Methods

3.1 Amplification of Gene of Interest and Digestion of Chosen Plasmid

1. Isolate DNA from your organism of interest (here, from multiple strains of *C. auris* that exhibit different Erg11 alleles).
2. Linearize pRS416 by digesting 2 μg of plasmid with SmaI restriction endonuclease in a 50 μL reaction. This will yield digested plasmid concentration of approximately 40 ng/μL.
3. Treat the digested plasmid with 2 μL (2 units) of SAP to help prevent recircularization.
4. Heat inactivate both enzymes by incubating the sample at 65 °C for 20 minutes (*see* **Note 6**). Digested and SAP-treated plasmid DNA can be stored at −20 °C for up to 6 months prior to use.
5. Design gap-repair primers that consist of approximately 20 nucleotides homologous to your gene of interest at the 3′ ends and approximately 50 nucleotides homologous to plasmid sequences flanking both sides of the SmaI cut site at the 5′ ends (Fig. 1).
6. PCR amplify your gene of interest (with or without regulatory regions) with your gap-repair primers and Phusion high-fidelity DNA polymerase. Set up four 50 μL reactions.
   (a) For a 50 μL reaction, combine (in order) 33.5 μL sterile water +10 μL 5× Phusion buffer +1 μL 10 mM dNTPs +2 μL 10 μM forward primer +2 μL 10 μM reverse primer +1 μL DNA (100–300 ng/μL) + 0.5 μL Phusion. The

PCR conditions are as follows: 98 °C for 30 s [98 °C for 15 s, calculated annealing temperature for 20 s, 72 °C for 15–30 s/kb] × 25, 72 °C 5 m.

(b) A negative control (no DNA) should be included to rule out contamination in the PCR reaction mixture.

7. Combine your PCR-amplified reactions. Set aside 5 μl of unpurified PCR and purify the remaining 145 μL of PCR product (Qiagen QIAquick kit can be used here). Elute in 30 μL to concentrate the DNA.
8. Gel electrophorese the 5 μL of unpurified and 1 μL of purified PCR product to confirm proper amplification at the expected size and no loss of product during purification. Add DNA loading dye to each and water for volume, as needed, prior to electrophoresis.
9. Determine the concentration of the purified PCR product using a Nanodrop or other spectrophotometer (*see* **Note 7**).

3.2 Co-Transformation of Competent S. cerevisiae *Cells with PCR and Digested Plasmid*

1. Combine 1.5 μL (~60 ng) of linearized pRS416 with 400–600 ng of purified PCR product in a microcentrifuge tube. Pipette up and down to mix.
2. Three control tubes should also be set up. Add 50–100 ng of uncut, empty plasmid to the positive control tube. Add 1.5 μL (~60 ng) of cut plasmid only to one negative control and 1.5 μL of water to another negative control (cells only, no plasmid or PCR DNA).
3. Add 50 μl of competent *S. cerevisiae* cells to each tube. Pipette up and down to mix and allow to sit on ice for 15 minutes. Do not vortex.
4. Add 450 μL of solution 3 from the Frozen EZ Yeast Transformation II kit. Manually invert the tube 5–6 times.
5. Incubate the tube on a rocker at 30 °C for 90 minutes (*see* **Note 8**). Manually invert the tube every ~15 minutes.
6. Spread 200 μL onto agar plates of selective medium (here: SC-ura). Allow plates to fully dry and then incubate at 30 °C for 3–4 days (*see* **Note 9**).

3.3 Colony PCR Screen of Transformants and Confirmation of Gene Sequence

1. Set up eight PCR tubes per transformation. Label the tubes #1–8 and add 10 μL of sterile water to each. Use a small pipette tip or loop to graze the top of a colony. Place these cells into the prelabeled #1 tube. Pick seven more colonies from the same plate adding cells from different colonies to the remaining tubes. For the whole plasmid transformation, screening three to five colonies is sufficient.

2. Perform a PCR using 1.5 μL of cell/water mix from each tube. The forward primer should be designed to adhere to the plasmid backbone upstream of the gene insert (e.g., T7) and reverse primer to adhere to the gene insert (Fig. 1) (*see* **Note 10**).
   (a) A negative control (no cells/DNA) should be included. A PCR control using whole plasmid and primers that adhere to the plasmid backbone can also be used (e.g., T7 and T3; Fig.1).
   (b) For 30 μl reactions, combine 10.5 μL sterile water +1.5 μL 10 uM forward primer +1.5 μL 10 uM reverse primer +15 μL 2× EmeraldAmp +1.5 μL cell/water mix. The PCR conditions are as follows: 95 °C for 2 m [95 °C for 40 s, calculated annealing temperature for 40 s, 72 °C for 1 m/kb] × 30, 72 °C 5 m.
3. Gel electrophorese 10 μL of each PCR product.
4. For clones/transformants that yield a product in the PCR screen, streak out ~2 μL of corresponding cell mixes from **step 3.3.1** onto SC-ura agar plates. Incubate at 30 °C for 2–3 days.
5. Grow up 2–3 clones from isolated colonies overnight in 3 ml of SC-ura broth.
6. Isolate DNA from each clone.
7. PCR amplify the entire gene insert region using backbone-specific primers that bracket your gene insert (e.g., T7 and T3) (Fig. 1). We also include a control that contains 0.5 μl of empty plasmid (should yield a significantly smaller product). Conditions similar to **step 3.3.2b** can be used.
8. Gel electrophorese to confirm expected fragment size and purify PCR products (Qiagen QIAquick kit can be used here).
9. Combine purified PCR product with sequencing primers, as needed for full gene insert coverage.
10. Obtain sequences from your lab's selected sequencing company.
11. Confirm gene sequence (*see* **Note 11**).
12. Perform antifungal susceptibility testing, and/or other relevant phenotypic assays, to determine the significance of the gene or allele(s) on the antifungal profile of the model organism, *S. cerevisiae* (Fig. 2).

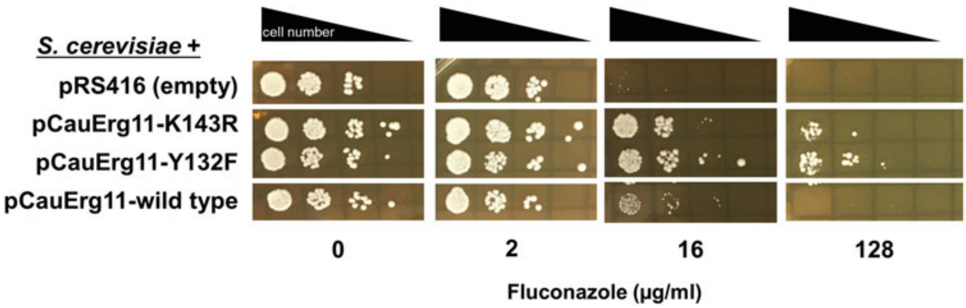

Fig. 2 Fluconazole spot assay of *S. cerevisiae* that express various alleles of plasmid-borne *ERG11* from *C. auris*. Numbers of cells spotted are tenfold reductions ranging from 1 × 10^4 (furthest left on each plate) to 10 cells (furthest right on each plate). Two alterations shown (K143R and Y132F) yielded decreased susceptibility toward fluconazole when compared to *S. cerevisiae* carrying an empty plasmid or a wild-type copy of Erg11. Additional strains and MICs obtained through broth microdilution can be found in [8]

4 Notes

1. To investigate multiple alleles of a gene, DNA from different strains that exhibit those alleles can be isolated and used separately in this protocol. Alternatively, one round of cloning with subsequent rounds of site-directed mutagenesis can be performed to obtain the desired set of plasmid constructs.
2. The *S. cerevisiae* host organism can be either haploid (BY4741 or BY4742) or diploid (BY4743 or INVSc1) depending on the ploidy of your organism of interest. Our labs use Zymo Research's Frozen EZ Yeast Transformation II kit to produce yeast competent cells. Alternatively, competent cells can be produced using a standard lithium-acetate protocol.
3. Other plasmids may be used here depending on your gene and strain of *S. cerevisiae*. We have used other pRS plasmids that contain other nutritional markers (e.g., *TRP1* or *HIS3*) or drug-selectable markers (e.g., nourseothricin resistance). Plasmids that contain specific promoters may be used if you wish to selectively or constitutively express your gene. In addition, "high copy" plasmids (2 µ *ori*) can be used to overexpress your gene.
4. The enzyme selected should cut the plasmid once (linearize), typically within the multiple cloning site. Blunt-end cutters are ideal to help prevent recircularization (in addition to the SAP treatment).
5. Any standard Taq polymerase, Taq buffer, and dNTPs (or other prepurchased DNA polymerase mastermix) can be used for screening transformants (colony PCR) and amplification of the gene insert for sequencing.

6. SAP is heat inactivated at 65 °C for 15 minutes and SmaI at 65 °C for 20 minutes. If you use an enzyme with a higher inactivation temperature, use the higher temperature to inactivate both enzymes.
7. Using this protocol, we typically obtain PCR concentrations of 300–600 ng/μL. The quality of starting genomic DNA will influence this yield. We have found that purified genomic DNA samples yield higher concentrations compared to DNA obtained through a brief boiling procedure, although the latter can still work if prepared fresh.
8. The rocking motion is important for proper transformation of yeast cells. A slow shaking/circular motion is less efficient. Certain drug resistance markers (e.g., *NAT1*) may require an outgrowth step.
9. Colonies should be obtained on the selective medium agar plates that contain the gap-repair and whole plasmid transformations, while plates containing the cut plasmid (no PCR DNA) or cells only should contain few to no colonies. If there are many colonies on the cut plasmid (without PCR DNA) control plate, this may indicate that the plasmid was not sufficiently digested and/or recircularized at a high rate. Repeat the enzyme digest at twice the length of time recommended by the supplier and ensure that the SAP treatment step is performed. Colonies or a lawn on the cells-only plate indicates contamination of the *S. cerevisiae* cells or the incorrect strain (does not have expected auxotrophy).
10. A product of 300–500 bp works well here. Do not discard the cell mixes. They will be utilized in a later step.
11. Sequencing software programs can be used here and/or free programs readily available online, such as chromatogram viewers, BLAST analysis suites, and sequence alignments to confirm sequences.

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